

Learning Outcomes

Wednesday, May 28, 2025 2:12 AM

CLO 1: Represent and solve problems in cylindrical and spherical coordinate systems

CLO 4: Perform operations on vectors in 2D and 3D Cartesian and curvilinear spaces

CLO6: Define and compute double and triple integrals over general regions

ULO 1.0: Coordinate Systems

ULO 1.1: Represent points and vectors in Cartesian, Cylindrical, and Spherical 3D spaces.

ULO 1.2: Convert between Cartesian and Cylindrical coordinates

ULO 1.3: Convert between Cartesian and Spherical Coordinates

ULO 3.1: Discuss the properties, calculation, and applications of the gradient, divergence, and curl.

Objectives:

- Transform coordinates between different systems
- Calculate the derivative matrix, Lamé coefficients, and Jacobian
- Compute gradient, divergence, and curl in different coordinate systems
- Use the Jacobian in change of variables for double and triple integrals